Section 3.3 — Quadratic Functions

Math 103: College Algebra

Solving Quadratic Equations

For each quadratic equation:

- If it is in standard form, solve by factoring.
- If it is in a form to allow it to be easily solved using the square root property, solve using the square root property.

1. $x^2 - 5x + 6 = 0$
2. $x^2 = 9$
3. $(x-5)^2 = 49$
4. $4x^2 + 4x - 15 = 0$
5. $3x^2 - 16x = 12$
6. $2(x-3)^2 + 4 = 6$

Graphing Quadratic Functions

For each of the following quadratic functions:

a. Find the x- and y-intercepts for the graph.

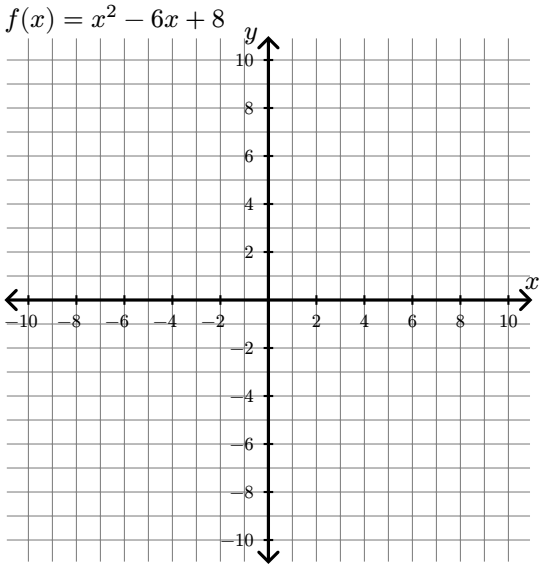
b. Using your x-intercepts, find the equation of the axis of symmetry and the coordinates of the vertex.

c. Create a table of x and y values that include at least the intercepts, the point with the same y-coordinate as the y-intercept, and two integer points on each side of the vertex.

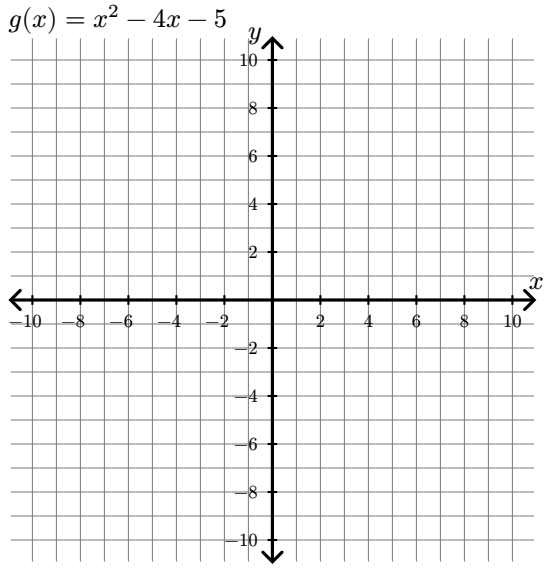
d. Sketch the axis of symmetry as a dashed line, and plot the vertex. (Estimate fractions if necessary.)

e. Graph the function in the space provided, plotting all appropriate points from your table.

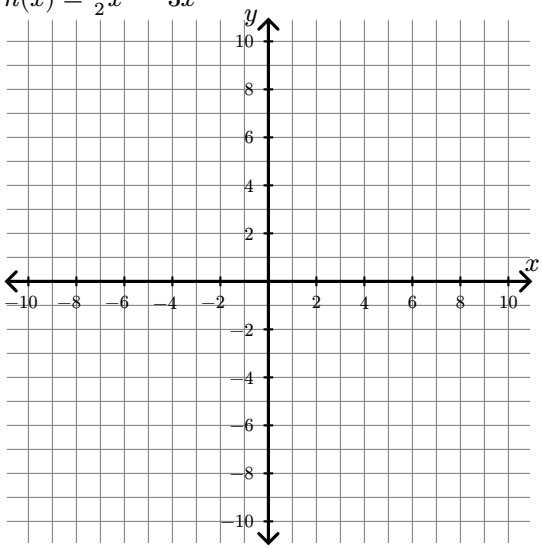
7. $f(x) = x^2 - 6x + 8$



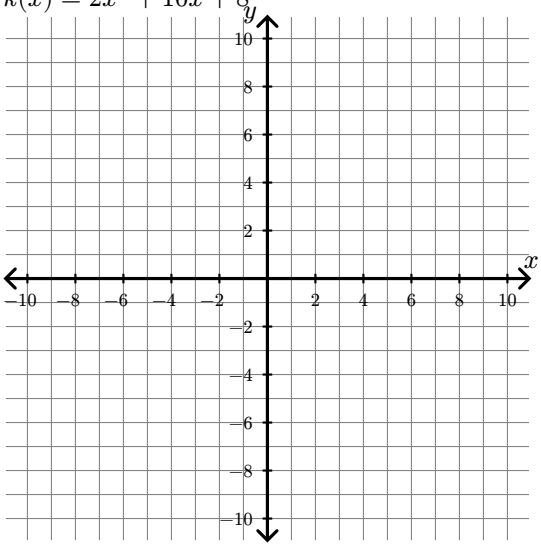
8. $g(x) = x^2 - 4x - 5$



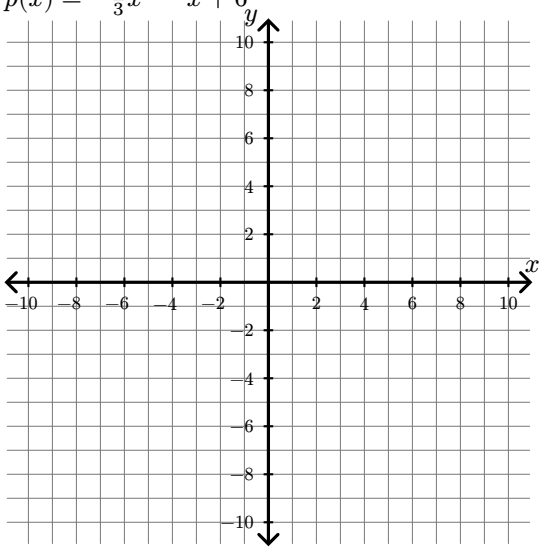
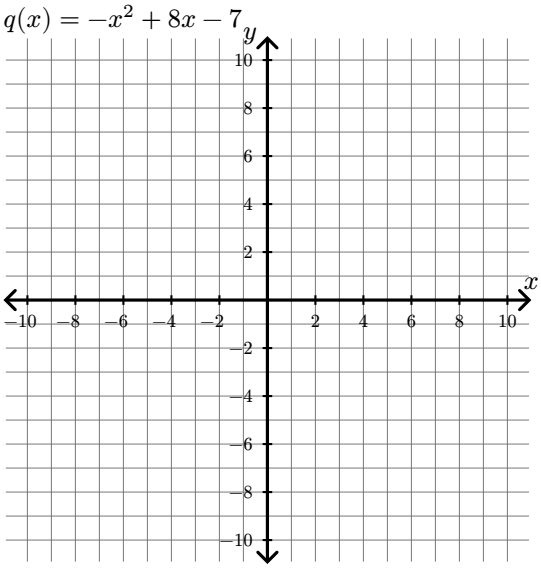
9. $h(x) = \frac{1}{2}x^2 - 3x$



10. $k(x) = 2x^2 + 10x + 8$



11. $p(x) = -\frac{1}{3}x^2 - x + 6$



12. $q(x) = -x^2 + 8x - 7$